

Generalized Bonferroni and Minimax Error Spending

Sample size savings relative to standard Bonferroni for heterogeneous endpoints

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GOAL & SETUP

Goal: Compare standard Bonferroni, proportionate spending, and minimax optimal spending while controlling familywise Type I and Type II error rates to optimize required sample size.

We consider the common setting where multiple endpoints are tested using the same sample, so the study size must satisfy all tests simultaneously.

For d tests with effect sizes $\delta_1, \dots, \delta_d$, the required sample size for test j is

$$n_j = \left(\frac{z_{\alpha_j} + z_{\beta_j}}{\delta_j} \right)^2 \quad N = \max_j n_j$$

Here, α_j and β_j are Type I and Type II error probabilities assigned to test and N is the overall sample size required to satisfy all tests.

Question: When does equal splitting waste sample size and how much can optimal spending recover?

ERROR SPENDING RULES

We compare three ways of allocating the total Type I and Type II error budgets across tests to optimize required sample size.

1. Standard Bonferroni: split error equally across all tests,

$$\alpha_j = \frac{\alpha}{d} \text{ and } \beta_j = \frac{\beta}{d}$$

2. Proportionate spending: allocate error according to test difficulty using constants C_α and C_β

$$\alpha_j = \Phi(-C_\alpha \delta_j), \quad \beta_j = \Phi(-C_\beta \delta_j)$$

3. Minimax optimal spending: choose α_j and β_j to minimize the largest required sample size

$$\min_{\{\alpha_j, \beta_j\}} \max_j n_j \text{ subject to } \sum_{j=1}^d \alpha_j = \alpha, \sum_{j=1}^d \beta_j = \beta$$

MINIMAX ALLOCATION SYSTEM

Our R function OptErrorSpending() solves for (C, N) from:

$$\alpha = \sum_{j=1}^d \Phi\left(-\frac{\delta_j^2 N + C}{2\delta_j \sqrt{N}}\right) \quad \beta = \sum_{j=1}^d \Phi\left(-\frac{\delta_j^2 N - C}{2\delta_j \sqrt{N}}\right)$$

Once C and N are determined, the value N represents the required overall sample size under minimax optimal error allocation, and the corresponding allocations α_j and β_j are then obtained:

$$\alpha_j = \Phi\left(-\frac{\delta_j^2 N + C}{2\delta_j \sqrt{N}}\right), \quad \beta_j = \Phi\left(-\frac{\delta_j^2 N - C}{2\delta_j \sqrt{N}}\right)$$

Key expectation

If all δ_j are equal, minimax collapses to equal splitting. If one test is harder (smaller δ_j), minimax should move more α and more β to that test.

If one test is harder (ie smaller δ_j), the minimax rule shifts more Type I error α and type II error β toward that test in order to balance required sample sizes across endpoints.

This redistribution should reduce the overall required sample size compared with equal splitting.

PREVIOUS ASYMPTOTIC RESULT

Earlier work showed that equal Bonferroni splitting introduces an unavoidable additive sample-size cost when multiple endpoints are tested simultaneously.

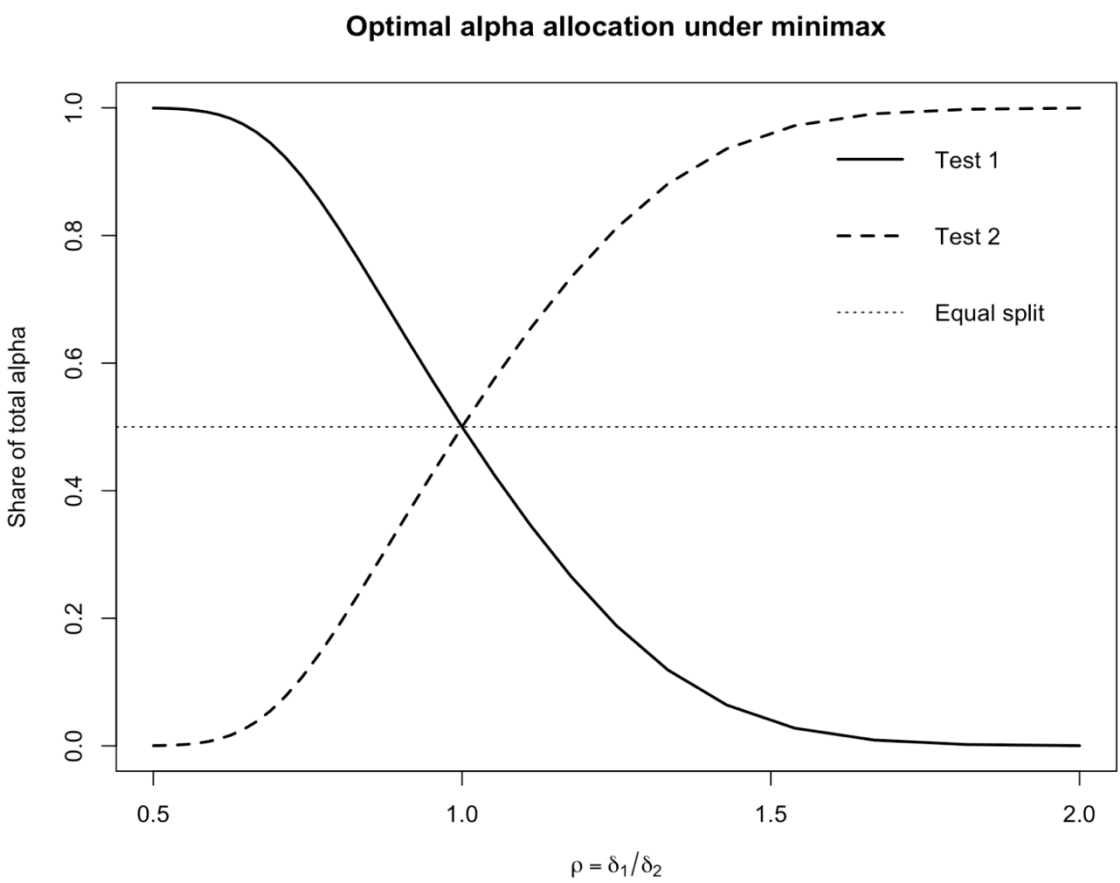
Specifically, as $\alpha, \beta \rightarrow 0$,

$$N_{\text{BONF}} - N_{\text{no-adjust}} \approx \frac{2 \log(k)}{\delta^2}$$

Although the relative cost of Bonferroni splitting vanishes as $\alpha \rightarrow 0$, this additive cost remains nonzero.

This motivates studying whether optimal error allocation can further reduce the required sample size when endpoint difficulties differ.

Figure 1. Optimal alpha allocation under minimax



Main message: the harder test receives a larger share of α .

What Figures 1 shows:

As effect sizes become unequal, the harder test receives a larger share of both α and β under the minimax rule. The optimal beta allocation under minimax graph produced the same results as the figure above.

NUMERICAL STUDY

We fix the total Type I and Type II error budgets α and β , vary the effect-size ratio $\rho = \delta_1/\delta_2$, and compare the overall required sample size under three error-spending rules:

1. Standard Bonferroni splitting,
2. Proportionate error spending,
3. Minimax optimal error spending

For each value of ρ , we compute the required overall sample size

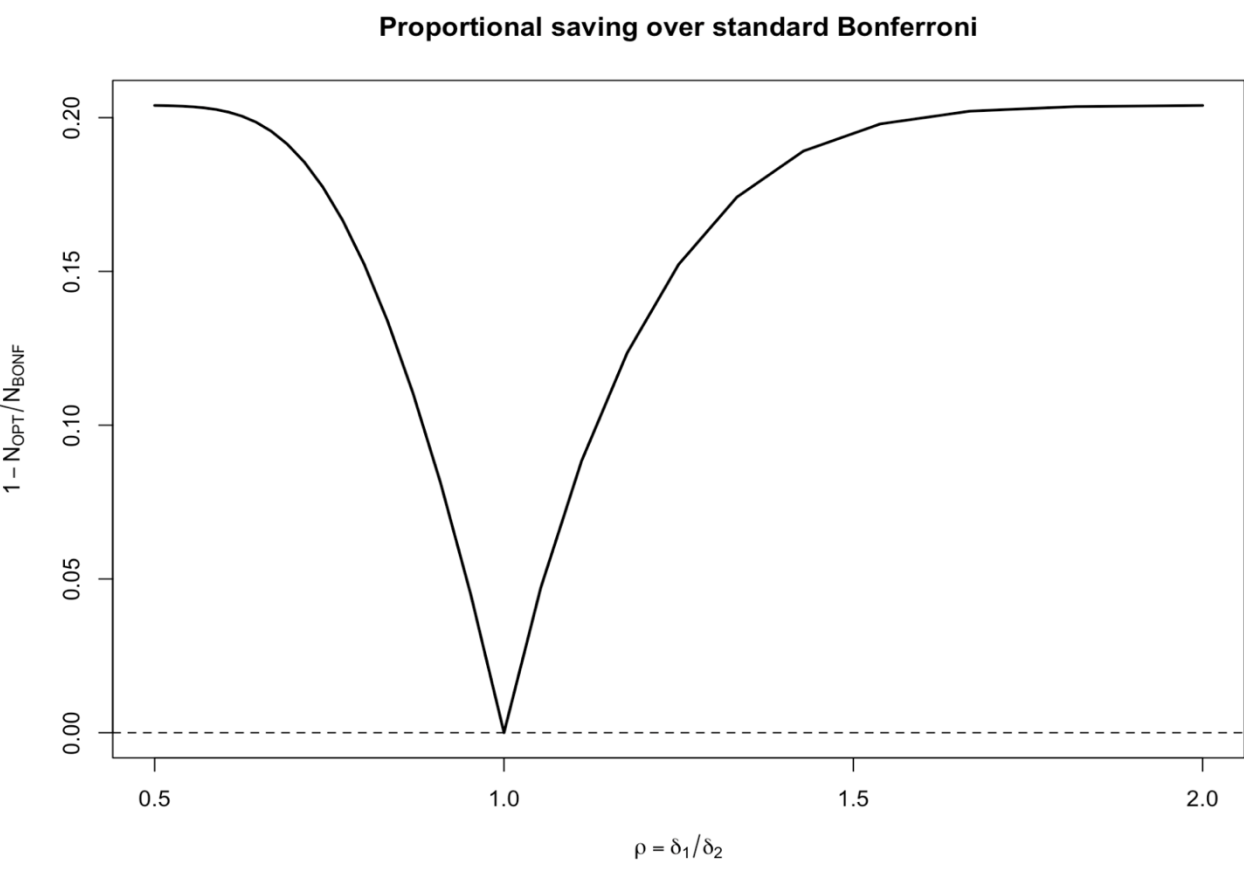
$$N = \max_j n_j$$

and examine how the allocation of error and the resulting sample-size savings change as the test difficulties become more unequal.

Primary diagnostics include:

- Allocation of Type I error (alpha shares)
- Allocation of type II error (beta shares)
- Proportional saving relative to standard Bonferroni
- Absolute sample-size saving
- Direct comparison of required sample size across methods

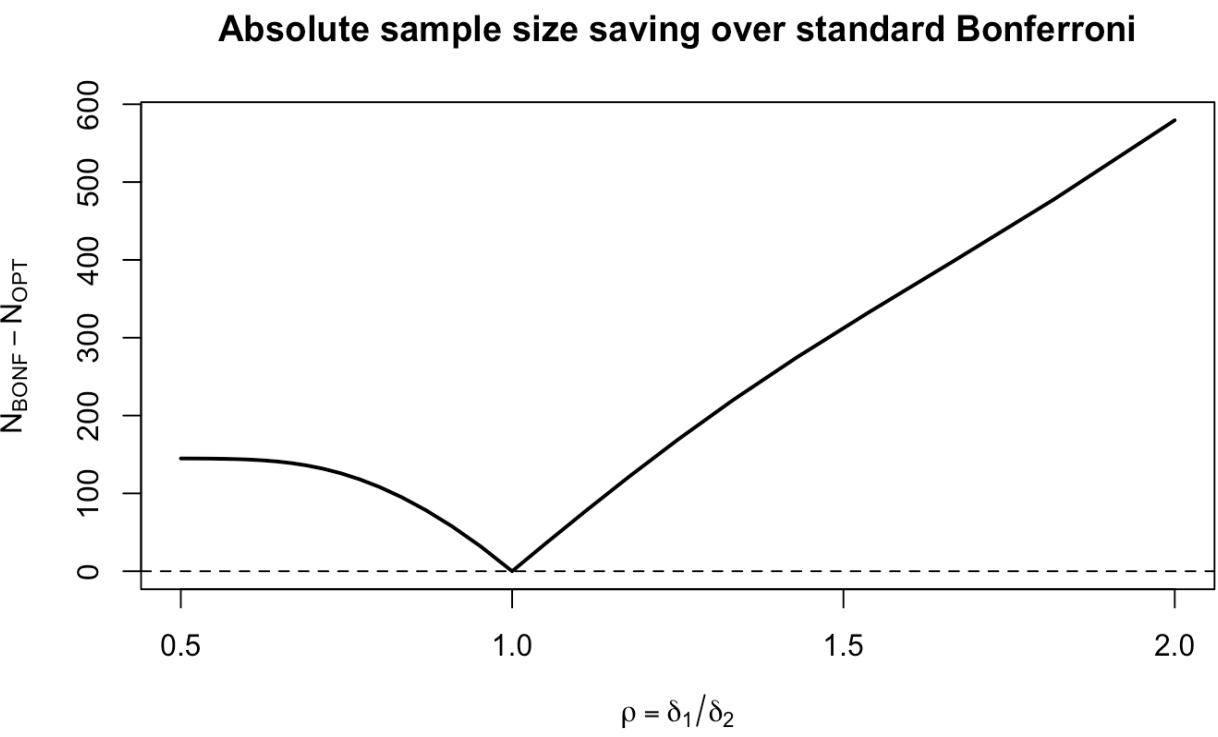
Figure 2. Proportional saving over standard Bonferroni



What Figure 2 Shows:

The proportional reduction $1 - N_{\text{OPT}}/N_{\text{BONF}}$ is largest when endpoint effect sizes differ most and vanishes when the effect sizes are equal.

Figure 3. Absolute sample-size saving



What Figure 3 Shows:

The difference $N_{\text{BONF}} - N_{\text{OPT}}$ increases with endpoint heterogeneity, showing the absolute number of sample-size units recovered by optimal allocation.

Table 1. Worked example comparison

Method	Overall N	Alpha 1	Alpa 2	Beta 1	Beta 2
Bonferroni	262.05	.025	.025	.0750	.0750
Proportionate	164.85	1.2e-05	.049988	.00343	.14656
Minimax	163.46	1.8e-04	.049821	.000429	.14957

What Table 1 shows:

Relative to standard Bonferroni splitting, both proportionate and minimax allocation concentrate Type I and Type II error on the more difficult endpoint, reducing the required overall sample size from 263 to approximately 164.

Minimax allocation achieves the smallest overall sample size while satisfying the same total Type I and Type II error constraints.

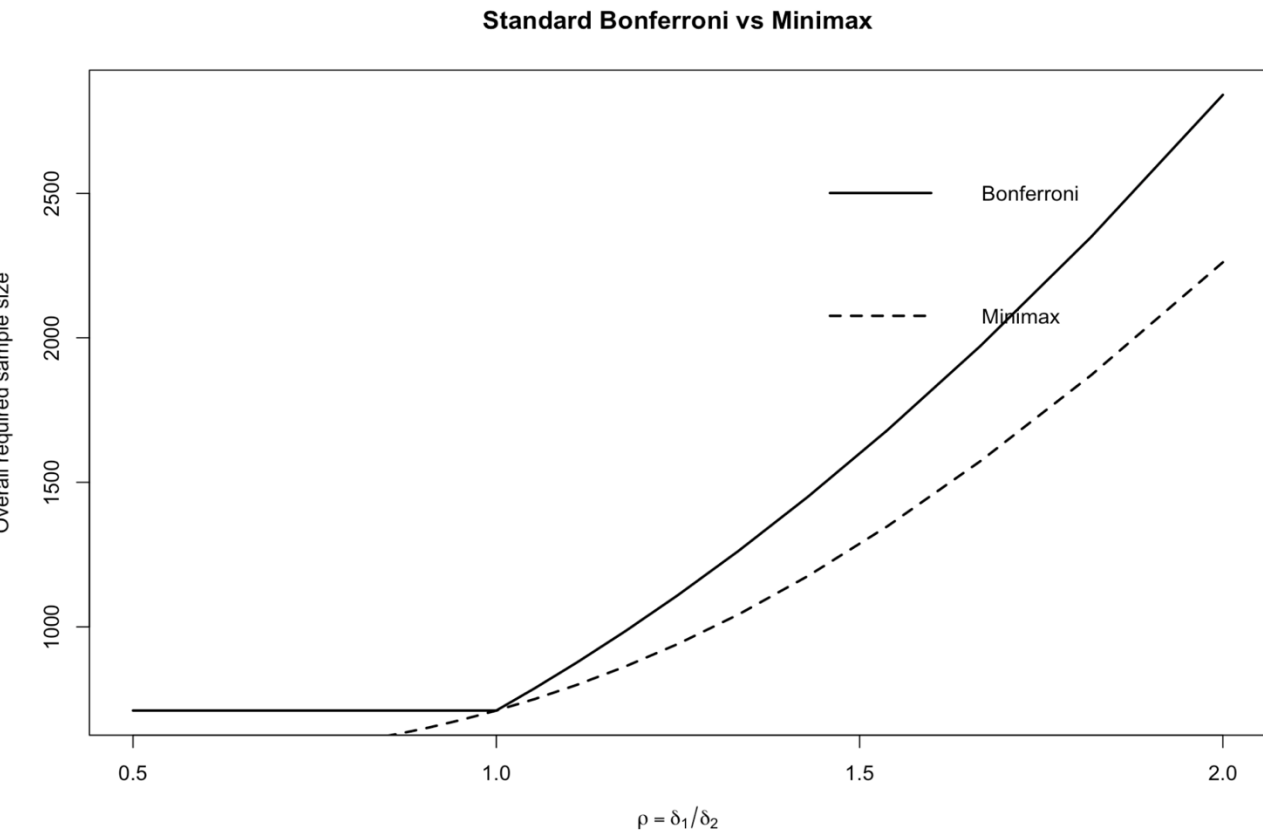
KEY RESULTS

- Equal Bonferroni splitting is not optimal when tests differ in difficulty (i.e., when effect sizes δ_j are unequal).

- The minimax allocation redistributes both Type I error (α) and Type II error (β) toward the harder test (with smaller δ_j)

- This reallocation reduces the overall required sample size, so the optimal sample size, N_{OPT} , is smaller than the Bonferroni sample size, N_{BONF} , whenever the effect sizes are unequal.

Figure 4. Standard Bonferroni vs Minimax



What Figure 4 Shows:

Minimax allocation reduces the required sample size relative to equal Bonferroni splitting whenever endpoint effect sizes differ.

INTERPRETATION

- At $\rho = 1$, the two methods are equal, so there is no advantage to reallocating error, confirming that equal splitting is optimal when effect sizes are equal

- As the effect sizes become more unequal, the smaller effect size determines the required overall sample size and receives a larger share of both Type I error (α) and Type II error (β) under the minimax rule

- As a result, the minimax method reduces the overall required sample size relative to standard Bonferroni splitting, with larger savings occurring as the heterogeneity between effect sizes increases.

- This confirms that optimal error allocation improves efficiency primarily when endpoints differ in difficulty.

FINAL CONCLUSIONS

- The minimax error allocation system was implemented numerically using Dr. Baron's R functions to determine optimal Type I and Type II error spending across tests.

- When effect sizes differ across endpoints, the minimax rule reallocates both α and β toward the more difficult test (with smaller δ_j)

- This optimal allocation reduces the overall required sample size relative to equal Bonferroni splitting, and the savings can be quantified both proportionally and in absolute sample size units.

- These results show that generalized Bonferroni / minimax allocation provides a more efficient testing strategy when endpoint difficulties are unequal, making it especially useful for clinical trials with heterogeneous endpoints

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